Game Theory Nine Notes

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1 Normalization of Payoff Matrices

1.1 Preservation of Nash Equilibrium

A fundamental property of bimatrix games is that Nash equilibria are invariant under affine transformations of payoff matrices. In particular, for any payoff matrix A, we may normalize entries to the interval $[0, 1]$ via the transformation

$$A' = \frac{A - \min(A)}{\max(A) - \min(A)}.$$

An identical transformation may be applied to the second player's matrix B. Note that the transformation applied to the first player's payoff matrix is not necessarily the same as the transformation applied to the second player's payoff matrix.

1.2 Why Normalization Matters

Normalization ensures that all payoffs are nonnegative and bounded. This is especially useful for:

- Vertex enumeration algorithms,
- The Lemke–Howson algorithm,

where negative values may distort geometric interpretations.

1.3 Example

Consider the payoff matrices

$$A = \begin{pmatrix} -2 & -7 & -3 \\ -3 & -2 & 1 \\ 1 & -3 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} -2 & -3 & 1 \\ 1 & -2 & -3 \\ -3 & 1 & -2 \end{pmatrix}.$$

After normalization, both matrices have entries in $[0, 1]$, which simplifies the geometry of the corresponding best-response polytopes used in Lemke–Howson.

2 BNN Dynamics and RK4 Error Analysis

2.1 Simple System: Two Strictly Dominant Strategies

Consider the game

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix}.$$

Each player has a strictly dominant second strategy.

Let

$$p_1 = (x_1, y_1), \quad p_2 = (x_2, y_2),$$

with

$$x_1 + y_1 = 1, \quad x_2 + y_2 = 1.$$

The BNN dynamics reduce to:

$$\dot{x}_1 = -2x_1^2, \quad \dot{y}_1 = 2x_1^2,$$

$$\dot{x}_2 = -2x_2^2, \quad \dot{y}_2 = 2x_2^2.$$

2.2 Exact Solution

This system admits a closed-form solution:

$$x_1(t) = \frac{x_1(0)}{1 + 2x_1(0)t}, \quad x_2(t) = \frac{x_2(0)}{1 + 2x_2(0)t}.$$

$$y_1(t) = \frac{y_1(0) + 2x_1(0)t}{1 + 2x_1(0)t}, \quad y_2(t) = \frac{y_2(0) + 2x_2(0)t}{1 + 2x_2(0)t}.$$

Thus, both inferior strategies converge to zero over time. This convergence is $O(t^{-1})$, which is quite slow

2.3 Numerical Results

The BNN modeled with RK4 converged to the actual solution, but seemingly at a low rate. To determine the order of convergence, the log(step size) vs log(error) plot will be recreated with far more evaluations in the 10^{-4} to 10^{-6} range.

In situations where the overall number of steps is limited, I found that increasing T (the time period considered for the ODE) was far more important than step size Δt, so long as the step size was small enough to ensure that the method remained in the stability. These results held across the more complicated games with one and then zero strictly dominant strategies.

A further area investigation will be using implicit methods to solve the BNN ODE system, as they have a larger domains of stability.

3 Fictitious Play with Continuous Strategies

3.1 Simultaneous Fictitious Play

For compact continuous strategy spaces and continuous ordinal potentials, simultaneous fictitious play converges (Danskin, 1954).

The simultaneous fictitious play for continuous strategies progresses through an iterative process:

Choose a random value x_0 for Player 1 and y_0 for Player 2

At each step:

$$x_{n+1} \in \arg\max_x \sum_{k=0}^{n-1} \Phi(x, y_k), \quad y_{n+1} \in \arg\max_y \sum_{k=0}^{n-1} \Phi(x_k, y).$$

Each player's strategy is then updated as a weighted sum of Dirac measures:

$$s^1_{n+1} = \sum_{k=0}^{n} \delta_{x_k} \quad s^2_{n+1} = \sum_{k=0}^{n} \delta_{y_k}$$

3.2 Alternating vs. Simultaneous Updates

There is no universal answer to which converges faster:

- Alternating FP has stronger convergence guarantees,
- Simultaneous FP better reflects parallel decision-making.

3.3 Connection to AI

This connects directly to:

- Multi-agent reinforcement learning,
- Neural fictitious self-play.

4 Fluid Dynamics and Game Theory

4.1 Mean Field Games

Mean field games relate strategic interaction to PDE systems.

A typical system involves:

- Hamilton–Jacobi equations (optimal control),
- Fokker–Planck equations (population evolution).

4.2 Connection to Fluid Dynamics

Luo and Song [2023] established the equivalence between Mean Field Games and a class of PDE systems closely realted to the compressible Navier Stokes equations.

Compressible Navier-Stokes equations

$$(\partial_t + v \cdot \nabla)v = -\rho^{-1}\nabla p + \rho^{-1}\nu\Delta v + \rho^{-1} f,$$

$$\partial_t \rho + \nabla(\rho v) = 0,$$

$$v(0,x) = v_0(x), \quad \rho(0,x) = \rho_0.$$

System Studied by Luo and Song

$$(\partial_t + v \cdot \nabla)v = -\nabla p[\rho] + \nu[\rho]\Delta v + f[\rho], \quad (t,x) \in [0,T] \times \mathbb{M}^3,$$

$$\partial_t \rho + \nabla \cdot (\rho v) + \nu[\rho]\Delta\rho = 0,$$

$$v(0,x) = v_0(x), \quad \rho(T,x) = \rho_T(x).$$

Mean Field Game PDE System

$$\frac{\partial u}{\partial t} - \Delta u + H(x, \nabla u) = V(x,m), \quad u\big|_{t=0} = V_0(x, m(x,0)),$$

$$\frac{\partial m}{\partial t} + \Delta m + \operatorname{div}\left(\frac{\partial H}{\partial p}(x, \nabla u)\, m\right) = 0, \quad m\big|_{t=T} = m_0,$$

$$m > 0, \quad \int m\, dx = 1 \quad \text{for all } t \in [0, T].$$

Studying the relationship between these three systems of PDEs will be the focus of next week's research.